Activity 6 Pre-Lab — Dehydration of Cyclohexanol

Organic Lab · Synthesis of cyclohexene + purification by simple distillation

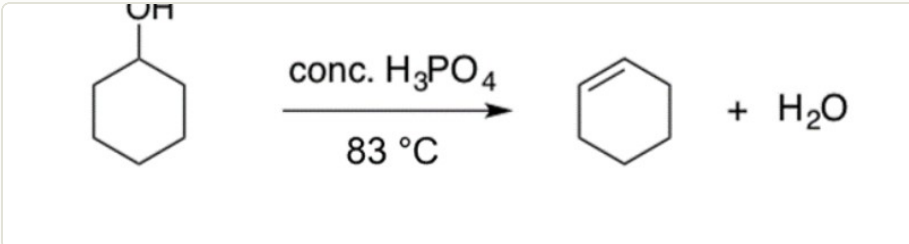
[Your name] & [partner]
[MM/DD/YY]

Black = write this in your notebook **Blue = context, don't copy it** Copy the header onto every page. Draw anything in a dashed box yourself.

Purpose

Synthesize **cyclohexene** by the acid-catalyzed **dehydration (E1)** of **cyclohexanol** using 85% phosphoric acid, isolate it by **simple distillation**, then check its purity/identity by TLC, the bromine test, and IR.

✍ **DRAW — reproduce this reaction in your notebook (from the procedure)**



mechanism = **E1**: protonate the –OH → lose H_2O (carbocation) → lose a β-H → cyclohexene. Also write the conditions (85% H_3PO_4, ≤83 °C, distill).

Chemical Reagents & Safety Table

CONTEXT ▸ Required: cyclohexanol, phosphoric acid, cyclohexene, sodium sulfate, sodium chloride. Typical SDS values — cross-check the SDS tile on Labflow.

Chemical	Role in lab	MW	Amount used	Physical properties	Hazards (GHS)
Cyclohexanol	Starting material (limiting)	100.16	10 mL (~9.6 g, ~0.096 mol)	Colorless liquid/waxy solid; bp 161 °C; d 0.96 g/mL; mp ~25 °C	H302 harmful if swallowed · H315 skin irritation · H332 harmful if inhaled · H335 respiratory irritation
Phosphoric acid (85%)	Acid catalyst / dehydrating agent	98.00	12 mL	Colorless viscous liquid; d 1.69 g/mL	H290 corrosive to metals · H314 causes severe skin burns & eye damage
Cyclohexene	Product	82.14	product (theor. ~0.096 mol)	Colorless liquid; bp 83 °C; d 0.81 g/mL; flash pt –12 °C	H225 highly flammable · H302 harmful if swallowed · H304 may be fatal if swallowed/enters airways · H315 · H411 aquatic tox
Sodium sulfate (anhydrous)	Drying agent	142.04	pea-sized	White solid; hygroscopic	No significant GHS hazards (drying agent) — still avoid dust/eyes
Sodium chloride (sat. aq.)	Brine wash (removes water)	58.44	2 × 10 mL	Colorless aqueous solution	No significant GHS hazards

FYI ▸ also used: 10% Na_2CO_3(aq) to neutralize phosphoric acid that distills over (fizzes → CO_2, vent often!); charcoal boiling chips (regular stones dissolve in H_3PO_4).

Activity 6 Pre-Lab (cont.) — dehydration of cyclohexanol [Your name] · [MM/DD/YY]

GRAB FIRST: 100-mL round-bottom (reactor) · 50-mL round-bottom (receiver) · simple-distillation head + condenser + thermometer adapter + thermometer · heating mantle + Variac · lab jack + clamps · ice-water bath · small separatory funnel · 25-mL Erlenmeyer · charcoal boiling chips.

Procedure Plan (left = your plan, right = fill in during lab)

Plan (write before lab)

Part 1 — Synthesis + distillation

1. Measure **~10 mL cyclohexanol** into the **100-mL RBF**; record exact volume.
2. Add **12 mL 85% H_3PO_4** + a few **charcoal boiling chips**, swirl to mix. H_3PO_4 is corrosive — gloves; follow instructor's handling.
3. Clamp the reactor to the bars; build a **simple-distillation** apparatus (still head → thermometer → condenser → 50-mL receiver in an **ice-water bath**). Heating mantle on a **raised lab jack** below. Jack must drop enough to slide the hot mantle out.
4. Instructor checks setup → heat with the **Variac**.
5. Keep the **vapor temp ≤ 83 °C** (adjust Variac / lower mantle). Record the steady distilling temp. 83 °C = cyclohexene bp; leaves cyclohexanol + H_3PO_4 behind.
6. Stop (lower jack, mantle off) when **5–10 mL** remain in the reactor. **Never distill to dryness.**
7. Neutralize: slowly add **10% Na_2CO_3(aq)** to the receiver in 5-mL portions, swirl till fizzing stops → transfer to sep funnel. Vent often in the hood — CO_2 builds pressure.
8. Wash the organic layer **2 × 10 mL saturated NaCl**; drain off the aqueous each time. Product = organic (top) layer.
9. Transfer product to a 25-mL Erlenmeyer; dry over a **pea-sized** amount of anhydrous **Na_2SO_4** until no longer cloudy.
10. Decant the clear cyclohexene into a pre-weighed flask; weigh → mass of product → **% yield**.

Tests (do all before discarding product)

11. **TLC** (30% EtOAc/hexanes) · **Bromine test** (Br_2 decolorizes → alkene) · **IR** (look for =C–H ~3050, C=C ~1650; loss of broad O–H).

✍ WRITE
% yield = (actual mol cyclohexene ÷ theoretical mol) × 100

Experiment Notes (in lab)

Exact cyclohexanol volume: ________ mL

Steady vapor temp: ________ °C

Distillation observations:

Workup — layers / venting / bubbling:

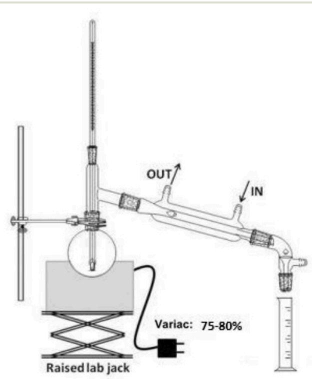
Product appearance (clear? turbid?):

Mass flask: ______ g **flask+product:** ______ g

Mass product: ______ g **% yield:** ______

Bromine test / TLC R_f / IR peaks:

Setup — draw this



simple distillation apparatus

CONTEXT (don't copy)
Why 83 °C cap? cyclohexene boils at 83 °C; cyclohexanol (161 °C) and H_3PO_4 stay behind, so the distillate is mostly pure product.

Why brine + Na_2SO_4? saturated NaCl pulls water out of the organic layer; anhydrous Na_2SO_4 then dries the last traces (cloudy → clear).

Clean-up & Conclusions

- Acidic pot residue → neutralize/dispose per instructor; organic waste for cyclohexene after tests; aqueous washes → aqueous waste; return glassware; do your bench task.
- **Conclusions** (~4 sentences after lab): your % yield and a reason it wasn't 100% (product left in pot, evaporation, incomplete rxn); what the bromine test + IR C=C confirmed; and how distillation purified the product.